Supplementary Material for:
Optimal Stratification of a Sampling Frame: A Comparative Study on Classical, Quantum, and Quantum-Inspired Approaches

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This document collects, in expanded form, the commercial and platform-survey detail that was condensed in the main paper (Sections 7 and 8) for reasons of length. It is intended as a companion reference, not as a standalone argument: all cross-references to sections, figures, tables, equations and bibliography entries refer to the main paper, which should be read first. Nothing in this document is required to follow the main paper's methodology or conclusions.

1 Extended mathematical derivations

This section gives, in full, several derivations that the main paper states as results only, for length. Notation follows the main paper throughout.

1.1 Full expansion of the one-hot penalty (Section 4.2)

Because $x_{hk}^2 = x_{hk}$ for binary variables, the one-hot penalty can be expanded as

$$\left(1 - \sum_{k=1}^{K} x_{hk}\right)^2 = 1 - \sum_{k=1}^{K} x_{hk} + 2 \sum_{1 \le k < l \le K} x_{hk} x_{hl}.$$

Substituting into $Q_{\text{pen}}(X) = \lambda \sum_h (1 - \sum_k x_{hk})^2$ and adding the dispersion term $Q_{\text{disp}}(X)$ gives, apart from the constant λL, the complete quadratic polynomial quoted in the main paper,

$$Q(X) = -\lambda \sum_{h=1}^{L} \sum_{k=1}^{K} x_{hk} + \sum_{k=1}^{K} \sum_{1 \le h < h' \le L} w_{hh'} x_{hk} x_{h'k} + 2\lambda \sum_{h=1}^{L} \sum_{1 \le k < l \le K} x_{hk} x_{hl} + \lambda L.$$

After stacking the LK binary indicators into a vector x, this is the standard QUBO form $\min_{x \in \{0,1\}^{LK}} x^T Q x$, where the diagonal elements of Q contain the linear coefficients and the off-diagonal elements contain both the within-stratum dissimilarities and the one-hot penalties.

1.2 QUBO-Ising equivalence (Section 4.4)

The Ising representation expresses the QUBO energy through spin variables $s_i \in \{-1, +1\}$ rather than binary variables $x_i \in \{0, 1\}$, with linear terms describing single-spin contributions (main effects, in statistical language) and quadratic terms describing pairwise interactions. The two representations are equivalent under $s_i = 1 - 2x_i$, equivalently $x_i = (1 - s_i)/2$: substituting this relation into the QUBO changes the coefficients and may add an irrelevant constant, but preserves the ordering of configurations and the set of minimisers, since the transformation is a strictly monotonic (in fact affine) reparameterisation applied identically to every configuration.

1.3 Pairwise identity connecting the surrogate to the Bethel objective (Section 4.3)

The identity

$$\sum_{h:z_h=k} N_h \left(\overline{Y}_{hj} - \overline{Y}_{kj}\right)^2 = \frac{1}{2N_k(z)} \sum_{h=1}^{L} \sum_{h'=1}^{L} N_h N_{h'} \left(\overline{Y}_{hj} - \overline{Y}_{h'j}\right)^2 x_{hk} x_{h'k}$$

follows from the standard decomposition of a weighted sum of squared deviations from the mean into a sum over all pairs: expanding the right-hand side's squared difference and using $\sum_h N_h x_{hk} \overline{Y}_{hj} = N_k(z)\overline{Y}_{kj}(z)$ and $\sum_h N_h x_{hk} = N_k(z)$ recovers the left-hand side after cancellation. For a fixed value of N_k, the numerator on the right-hand side is quadratic in the assignment indicators; because $N_k(z)$ itself varies with the candidate partition, dropping the factor $1/\{2N_k(z)\}$ to obtain a valid QUBO changes the criterion and can alter the ranking of partitions, especially when final-stratum sizes differ substantially.

1.4 Exact expectation of the implemented IBM Hamiltonian (Section 4.6)

Dropping the identity term from the cost Hamiltonian $H_C = \sum_{h<h'} w_{hh'} \delta(z_h, z_{h'})$ gives the operator actually implemented,

$$H_C^{\text{impl}} = H_C - \tfrac{1}{4} \sum_{h<h'} w_{hh'} I,$$

a constant shift lowering every eigenvalue uniformly. For uniformly random labels with $K = 4$, $P(z_h = z_{h'}) = 1/4$ for any pair $h \neq h'$, so $E[\delta(z_h, z_{h'})] = 1/4$ under the untranslated Hamiltonian; after the identity term is removed, the expected cost of a uniform distribution over the four labels is therefore exactly zero,

$$E\left[H_C^{\text{impl}}\right] = \sum_{h<h'} w_{hh'} \left(\tfrac{1}{4} - \tfrac{1}{4}\right) = 0,$$

by construction, not as an empirical coincidence requiring a separate explanation such as an offset or scaling error. The near-zero measured energy on the real IBM run is therefore exactly what the implemented Hamiltonian predicts for a broad, near-uniform output distribution; the evidence that the run failed to optimise lies in the absence of any energy improvement over the two iterations and in the diffuse, non-repeating bitstring distribution actually observed, not in the zero energy value taken on its own.

1.5 QAOA circuit structure and run parameters (Section 4.6)

QAOA searches the cost landscape through a parameterised quantum circuit. The circuit starts from a superposition of possible binary labels and alternates two operations: evolution under the cost Hamiltonian, which assigns lower energy to assignments with lower surrogate cost, and evolution under a mixing Hamiltonian, which allows the circuit to explore alternative assignments. For a circuit with p layers, these operations are repeated p times, their duration controlled by a set of variational parameters; the quantum processor prepares and measures the corresponding state, while a classical optimiser (COBYLA in this study) updates the parameters to reduce the estimated expected energy.

The real hardware run used the ibm_kingston backend (Heron r2, 156 qubits), shown in Figure 2 of the main paper. The exact QAOA layer count p and the number of shots per circuit evaluation were not recorded in a form that can be reported precisely here; the final partition was the lowest-cost bitstring observed among those sampled during the two COBYLA iterations. This gap affects only the precise reproducibility of this one run, not the conclusions drawn from it, which rest on the measured circuit depth and gate count rather than on these parameter values.

1.6 D-Wave annealing schedule (Section 4.5)

The search begins from the ground state of a simple transverse-field Hamiltonian H_0 and evolves according to the time-dependent Hamiltonian

$$H(t) = A(t) H_0 + B(t) H_P, \quad 0 \leq t \leq T,$$

where t denotes the progression of the annealing process. At the beginning, the transverse-field term dominates and allows the system to explore alternative configurations; toward the end, the problem Hamiltonian H_P dominates, so lower-energy configurations correspond to better values of the QUBO objective. A further implementation step is required because the logical Ising model is dense, whereas the QPU has a fixed sparse connectivity graph: minor embedding represents a logical variable by a chain of connected physical qubits, so the number of physical qubits can be substantially larger than the number of logical variables, and this can affect the quality of the search through chain length, chain strength, broken chains, coefficient scaling and hardware noise.

2 Extended discussion: what paid access would change

This section expands Section 7 of the main paper with the commercial and quota detail omitted there.

2.1 D-Wave

On the free plan the QPU solvers were not authorised at all, which is why the D-Wave QUBO was solved in this study with classical simulated annealing. D-Wave's hybrid solvers, available on paid plans, accept problems with up to a million variables and would swallow the original 164-stratum frame without difficulty; but they operate by classically decomposing the problem and invoking the QPU only on sub-problems, so the genuinely quantum share of the computation shrinks accordingly, and one is largely measuring a classical solver again.

2.2 IBM

The IBM free plan grants ten minutes of QPU time in a rolling twenty-eight-day window and forbids session mode, which is why the variational loop in this study had to be cut to two COBYLA iterations, each queued as an independent job. A paid plan removes exactly these restrictions: it provides far more QPU time, dedicated sessions that eliminate the queue between iterations, and access to error-mitigation and error-suppression techniques, so that the variational loop could run to convergence with dozens or hundreds of iterations and far more shots per circuit. As discussed in the main paper, this would not remove the circuit-depth limitation, which is physical rather than contractual.

2.3 Dirac-3: the 949 versus 954 discrepancy

The free tier rejects any degree-two problem with more than one hundred decision variables, which at $K = 4$ caps the frame at twenty-five atomic strata. The physical device is bounded instead by its total qudit-level capacity. QCi's own documentation is not perfectly consistent on this figure: the Dirac-3 Developer Beginner Guide states 949, while the eqc-direct integer-solver documentation states 954 and gives, as a worked example, a maximum of 477 binary variables, consistent with 954 rather than 949 (which would give 474). Neither source was the specific endpoint queried for the free-tier rejection reported in the main paper, which returned only the separate 100-variable limit. Lacking a documented figure for the exact endpoint relevant to this study, the main paper adopts the eqc-direct figure of 954 as the more specific and more recently verified of the two, giving roughly 119 atomic strata at $K = 4$ under full paid access.

2.4 Trapped ions: access and cost

Access to trapped-ion hardware (IonQ, Quantinuum) is through Amazon Braket or Microsoft Azure Quantum, both pay-per-use. Microsoft has historically advertised an automatic credit of USD 500 per participating hardware provider, and up to USD 10,000 through its Research Credits programme; these specific figures do not appear on the current Azure Quantum pricing page, and their continued availability, amount and eligibility depend on the specific provider, region, subscription plan and time period, change without notice, and should be re-verified directly with Microsoft before being relied upon. A word of caution on cost: IonQ bills per gate-shot, so cost depends on the number of shots and on whether error mitigation is enabled, neither of which is fixed here; no explicit cost estimate is offered for the $K = 4$ circuit, with its roughly two thousand two-qubit gates, beyond noting that it is the more expensive of the two options by a wide margin. The $K = 2$ encoding, at twenty qubits and 380 two-qubit gates on a fully connected layout, is the cheaper test and the one recommended in the main paper as the first, minimal-cost step, though its exact cost was likewise not computed here.

3 Platforms considered but not used

The four solvers used in the main study were chosen because they were freely accessible, not because they are the only options. This section surveys the platforms that were considered but not used, and explains why.

3.1 Google: effectively closed

The obvious omission is Google. As of access in July 2026, its hardware is not generally available: the Quantum Computing Service has remained a limited preview granting time to selected research partners rather than open self-service, so one cannot simply register and run circuits on the Willow processor, in contrast with IBM's open cloud (Google Quantum AI, 2026). Google had opened a Willow Early Access Program inviting research proposals; that route was by competitive selection, with a proposal deadline of 15 May 2026, and the programme page now indicates that selected applicants have been notified, at least one such selection (a King's College London team) having been publicly announced in late May 2026. The programme was, in any case, aimed at experiments designed specifically to probe the device rather than at applied optimisation studies, and no application was made for the present study. This is a time-sensitive, commercial detail that should be re-verified against Google's current documentation before being relied upon; for the purposes of a study like this one, at the time of access, Google was not an option.

3.2 Other dense-QUBO machines, and why Amplify AE was enough

The family of machines purpose-built for densely connected QUBOs is broader than the single member used in the main study. The Fujitsu Digital Annealer solves fully connected QUBO problems on application-specific CMOS hardware, with an annealing algorithm using parallel-trial and dynamic-escape mechanisms (Aramon et al., 2019), and its fully connected architecture is aimed precisely at problems that are hard for both conventional computers and quantum annealers. Toshiba's SQBM+, built on the Simulated Bifurcation algorithm (Goto et al., 2019), and NEC's Vector Annealing service occupy the same niche.

Neither was run, for a practical and a scientific reason. Practically, neither is free: SQBM+ requires a separate subscription with its own endpoint, and the Digital Annealer a Fujitsu licence, whereas Amplify AE is available at no cost. Scientifically, they would add comparatively little on their own, because Amplify AE has already provided the key measurement: the best-known QUBO solution found when quantum-specific hardware limitations are absent and a strong classical heuristic is used, corroborated by one further independent simulated-annealing search. A second or third dense-QUBO machine could provide an additional independent comparison with this best-known result, but would be unlikely to change the qualitative picture already established.

One product deserves a word only to set it aside. Fixstars also markets Amplify SE, the Scheduling Engine, which is sometimes mistaken for a variant of AE. It is a different tool entirely: a domain-specific solver for scheduling problems, in which the user declares jobs, machines and tasks with their processing times, and the engine minimises the makespan. It deliberately requires no mathematical formulation from the user and does not accept a QUBO at all. Optimal stratification is not a scheduling problem, so Amplify SE has no bearing on it.

3.3 What would not help

Two other families were not pursued, for reasons specific to the present formulation rather than general unsuitability. Neutral-atom machines, such as QuEra's Aquila and Pasqal's Fresnel, are analog devices whose natural problem is the maximum independent set on unit-disk graphs; they were not pursued because the present dense clustering formulation does not map naturally to the native analog problem classes available on those platforms without substantial reformulation. The remaining gate-based superconducting platforms, Rigetti and IQM, share the same sparse fixed-layout connectivity that contributed to the IBM failure; they would not isolate the role of connectivity as cleanly as an all-to-all trapped-ion experiment, because they also rely on sparse fixed-connectivity layouts, though they could still differ from IBM in fidelity, native gate set, compilation and calibration, so a similar outcome is expected rather than certain.

3.4 Summary table

Platform	Type	Status	Reason
Fixstars Amplify AE	GPU Ising engine	USED (n = 160)	Free Basic plan; 8,192-bit fully-connected cap, our 80 variables well within it
D-Wave QPU	Quantum annealing	NOT USED	Not authorised on the free plan
D-Wave (Ocean classical SA)	Classical SA, same pipeline	USED (n = 162)	Ran in place of the QPU, on the free plan
IBM Quantum	Gate-based, sparse	USED (n = 262)	Severely truncated run at depth ~4,900
QCi Dirac-3	Photonic, all-to-all	USED (n = 286)	Did not reach the best-known QUBO value found in this study
Quantinuum / IonQ	Trapped ion, all-to-all	NOT USED - worth running	No SWAP overhead; paid, credits available
Fujitsu DA / Toshiba SQBM+	Dense-QUBO hardware	NOT USED - redundant	Could provide an additional independent comparison with the best-known result
Google Willow	Superconducting	NOT AVAILABLE	Closed; access by proposal only
Rigetti / IQM	Superconducting, sparse	NOT USED - less diagnostic for connectivity	Same sparse fixed-layout as IBM; would not isolate connectivity as cleanly as an all-to-all device
QuEra / Pasqal	Neutral atom, analog	NOT USED - requires substantial reformulation	Native analog problem class does not match the present dense clustering formulation
Fixstars Amplify SE	Scheduling engine	NOT APPLICABLE	Not a QUBO solver at all

Table 1: Summary of platforms considered in this study.

References

Aramon, M., Rosenberg, G., Valiante, E., Miyazawa, T., Tamura, H., and Katzgraber, H. G. (2019). Physics-inspired optimization for quadratic unconstrained problems using a digital annealer. *Frontiers in Physics*, 7:48.

Google Quantum AI (2026). Willow early access program. https://quantumai.google. Accessed July 2026; programme details are time-sensitive.

Goto, H., Tatsumura, K., and Dixon, A. R. (2019). Combinatorial optimization by simulating adiabatic bifurcations in nonlinear Hamiltonian systems. *Science Advances*, 5(4):eaav2372.